

Diagonal Unloading Algorithm for Mini-Drone Localization Using Circular Microphone Array

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Abstract—The objective of this work is to operationalize and validate an acoustic platform designed for mini drones localization. This is achieved by deploying a microphone array arranged in a circular configuration. The core methodology involves implementing the diagonal unloading algorithm for direction of arrival estimation, that allows accurate drone localization. Various simulations are conducted to evaluate the algorithm's performance using Signal-to-Noise Ratio (SNR), the number of snapshots used for covariance matrix estimation and the reverberation time RT60. Comparative analysis is performed against considered algorithms such as MUSIC, MVDR, SRP and SRP-PHAT. Remarkably, the obtained results demonstrated that the diagonal unloading algorithm achieved comparable outcomes to the MUSIC algorithm while requiring less computational complexity. The validation process is further strengthened by experimental results obtained through practical implementation.

Index Terms—Mini-drone, Diagonal unloading algorithm, Acoustic localisation, Circular microphone array.

I. INTRODUCTION

In recent years, the proliferation of mini-drones, also referred to as mini Unmanned Aerial Vehicles (UAVs), has transformed the landscape of unmanned aviation. These diminutive aircraft, devoid of an onboard human pilot, are capable of remote, semi-autonomous, or fully autonomous operations. Due to their modest dimensions and limited endurance, typically ranging from 10 to 30 minutes [1], they have garnered significant attention across various sectors.

The surge in the utilization of small UAVs within the public domain has regrettably paralleled an upsurge in potential security threats. Instances of malevolent activities involving mini-drones, such as their exploitation for drug and weapon transportation, espionage endeavors, and even acts of terrorism have been noted. In response to these escalating challenges, the imperative to conceive and deploy effective countermeasures has become increasingly apparent. This is particularly crucial in safeguarding critical infrastructures and individuals from potential harm [2].

Within this context, the focus of this research lies in the development of new techniques for the localization of mini-drones. In fact, addressing these concerns require harnessing advancements in signal processing technologies. In this context, previous works proposed methods that exploit the distinct acoustic signatures emitted by UAVs. A significant contribution to this field is encapsulated within the paper [3], wherein the authors introduce a bio-inspired signal processing technique. This approach capitalizes on the auditory characteristics of mini-drones, employing a microphone array for detection and achieving exceptional accuracy in localization. Building upon this foundation, the study outlined in [4] delves further into the synergy between microphone arrays and real-time signal processing. The methodology leverages the acoustic sound generated by UAV motorization and aerodynamic phenomena, extracting intricate harmonics through innovative signal processing techniques. These harmonics serve as the basis for precise localization via Kalman filtering.

The efficiency of such endeavors is corroborated by the findings documented in [5]. This article presents a holistic algorithm that seamlessly merges acoustic signal analysis with concurrent neural networks (CoNNs). The algorithm's components encompass sophisticated spectrogram decomposition and adaptive filtering mechanisms, enabling meticulous analysis and processing of acoustic signals. Validation of its capabilities employs dedicated spiral microphone arrays, thereby substantiating its efficiency in detection and tracking.

Recently, it was shown that the Diagonal Unloading technique (DU) [6][7] can be used as an efficient technique to locate sound sources, achieving high resolution performance with low computational complexity. The idea of this algorithm is to unload the covariance matrix from the signal subspace, then exploit the orthogonal property between the signal and noise subspaces to obtain comparable results with high resolution algorithms. An improved version of this algorithm is given in [8], where the authors use the power method technique to estimate the largest eigen value of the covariance matrix to

improve its performance. In this work, we comprehensively examine the algorithm's performance for localizing mini-drones and contrast it with other algorithms. Our objective is to evaluate its effectiveness and suitability within a practical experimental setup.

The rest of this work is structured as follows. In section II, we introduce the conventional signal model for a circular array and we provide an overview of the Diagonal Unloading algorithm. In section III, we discuss the different comparisons and numerical results. The experimental results are presented in section IV. Section V concludes the paper.

II. SIGNAL MODEL AND PROBLEM FORMULATION

Suppose that a K acoustic drone signals $s_k(t, f), k = 1, 2, \dots, K$ propagate over a Uniform Circular Array (UCA) consisting of M omnidirectional elements with a radius r . The structure of the UCA is depicted in Figure (1), with the origin of the coordinate system as the reference point. The azimuth angle of the source θ , ranges from 0 to 2π , while the elevation angle, ϕ , ranges from 0 to $\pi/2$. These angles are measured from the x to y axes and from x to z axes of a spherical coordinate system, respectively. Consequently, the received signal by the m^{th} sensor is given by the equation (1).

$$x_m(t, f) = \sum_{k=1}^K e^{-j2\pi f \tau_m(\theta_k, \phi_k)} s_k(t, f) + n_m(t, f) \quad (1)$$

Where, f is the frequency bin, $\tau_m(\theta_k, \phi_k)$ is the propagation delay between sensor m and the origin. The noise $n_m(t, f)$ is assumed to be spatially white, following a Gaussian distribution with zero mean and a variance of σ^2 . The steering vector corresponding to the direction $\Omega_k = [\theta_k, \phi_k]$ is given by the equation (2).

$$\mathbf{a}(\mathbf{f}, \Omega_k) = \mathbf{a}(\mathbf{f}, \theta_k, \phi_k) = \begin{pmatrix} e^{-j2\pi r f \cos(\theta_k - \Phi_1) \cos(\phi_k)/c} \\ e^{-j2\pi r f \cos(\theta_k - \Phi_2) \cos(\phi_k)/c} \\ \vdots \\ e^{-j2\pi r f \cos(\theta_k - \Phi_M) \cos(\phi_k)/c} \end{pmatrix} \quad (2)$$

Where, $\Phi_i = \frac{2\pi i}{M}$ depicts the angular positions of the microphone array elements and c is the sound velocity. We can put all the received signals $x_m[t], m = 1, 2, \dots, M$ in a matrix X with the model given by the equation (3):

$$X(t, f) = A(f, \Omega)S(t, f) + N(t, f) \quad (3)$$

Where, $A(f, \Omega) = [a(f, \Omega_1) \ a(f, \Omega_2) \cdots a(f, \Omega_K)]$ is the steering matrix,

$S(t, f) = [s_1(t, f) \ s_2(t, f) \cdots s_K(t, f)]^T$ the source signals vector and $N(t, f) = [n_1(t, f) \ n_2(t, f) \cdots n_M(t, f)]^T$ the vector associated to the noise.

Given the assumption of spatially and temporally white noise that is uncorrelated with the sources, and considering

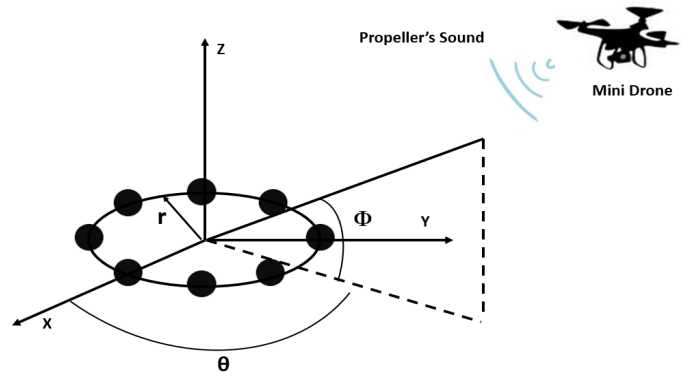


Fig. 1. Uniform Circular Array .

that the sources themselves are uncorrelated, we can derive the following covariance matrix:

$$\begin{aligned} R_{xx}(f) &= E[X(t, f)X(t, f)^H] \\ &= A(f, \Omega)R_{ss}(f)A(f, \Omega)^H + \sigma_n^2 I \\ &= A(f, \Omega) \begin{pmatrix} P_1^2 & & & \\ & P_2^2 & & \\ & & \ddots & \\ & & & P_K^2 \end{pmatrix} A(f, \Omega)^H + \sigma_n^2 I \end{aligned} \quad (4)$$

In practical scenarios, it is only feasible to obtain the spatial covariance matrix $R_{xx}(f)$ from a limited number of samples. As a result, the spatial covariance matrix is estimated using the maximum likelihood estimation, as shown in Equation 5.

$$\hat{R}_{xx}(f) = \frac{1}{T} \sum_{i=1}^T X(i, f)X^H(i, f) \quad (5)$$

Here, T represents the number of snapshots used in the formation of the spatial covariance matrix.

1) *Direction finding using diagonal unloading beamformer:* Diagonal unloading beamformer is an efficient algorithm which offers a comparable performance to high resolution algorithms such as MVDR and MUSIC with a low computational complexity. It has been shown in [8] that the DU spectrum can be expressed as follows (equation 6).

$$P_{DU}(f, \Omega) = \frac{1}{a(f, \Omega)^H (\lambda I - R_{xx}(f)) a(f, \Omega)} \quad (6)$$

Where, λ is the largest eigen value corresponding to subspace of the most powerful incident source, it can be obtained by the power method algorithm given by algorithm 1.

By employing the norm transform frequency fusion technique proposed in [9], it becomes possible to combine the various contributions and acquire a comprehensive pseudo-spectrum representation encompassing the entire frequency spectrum. This process yields a function that provides broadband pseudo-spectrum information given by equation (7).

Algorithm 1 Power method**Input:** R **Parameters:** Nt, ϵ **Initialization:** V_{old} **for** $n = 1$ **to** Nt **do** $V \leftarrow RV_{old}$ $\lambda \leftarrow \|V\|$ $V_{new} \leftarrow \frac{V}{\|V\|}$ **if** $\|V_{new} - V_{old}\| < \epsilon$ **then** **break** **else** $V_{old} \leftarrow V_{new}$ **Output:** λ

$$P_{DU}(\Omega) = \sum_{f=f_{\min}}^{f_{\max}} \left(\frac{P_{DU}(f, \Omega)}{\max(|P_{DU}(f, :)|)} \right) \quad (7)$$

Then finally, the direction of arrival of the most powerful source can be obtained by finding the maximum of the DU spectrum as shown in the equation (8).

$$\Omega_{estimate} = \arg \max_{\Omega} P_{DU}(\Omega) \quad (8)$$

III. NUMERICAL RESULTS

In this section, numerous simulations are conducted to evaluate the performance of the DU algorithm in comparison to other algorithms such as SRP, SRPPHAT [10], MVDR with diagonal loading [11] and MUSIC [12].

During the evaluation of the different estimation algorithms, we used Monte Carlo simulations to compute the Root Mean Squared Error (RMSE), which served as the performance measure. The RMSE was assessed considering the Signal-to-Noise Ratio (SNR) and the number of snapshots employed for covariance matrix estimation given by the Equation 5. Moreover, we compared the performance of estimating the direction of arrival in both azimuth and elevation. Equation 9 gives the formula used to compute the RMSE.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\theta_i - \hat{\theta}_i)^2 + (\phi_i - \hat{\phi}_i)^2} \quad (9)$$

Where, $\hat{\theta}_i$ and $\hat{\phi}_i$ represent the estimated value of the azimuth and elevation angles, respectively. The RMSEs are computed by averaging the results from $N = 200$ Monte Carlo trials where in each trial, we simulate an acoustic room using image source method [13], and we used the dataset in [14] to generate an acoustic drone signal in a random direction in elevation and azimuth. For the diagonal loading in the MVDR algorithm, we determined a parameter value of 0.1, the tolerable error for the power method algorithm is set to $\epsilon = 10^{-3}$, and the number of iterations is $Nt = 200$. We tested these algorithms on a circular array of 8 uniformly spaced microphones with an FFT window length of 1024 points. Figure 2 plots the RMSEs of the algorithms as a function

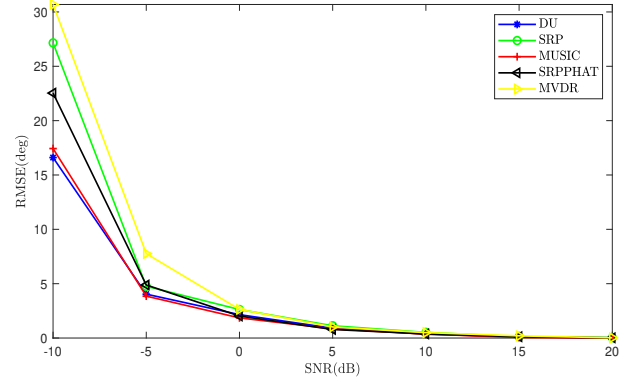


Fig. 2. RMSE performance versus SNR

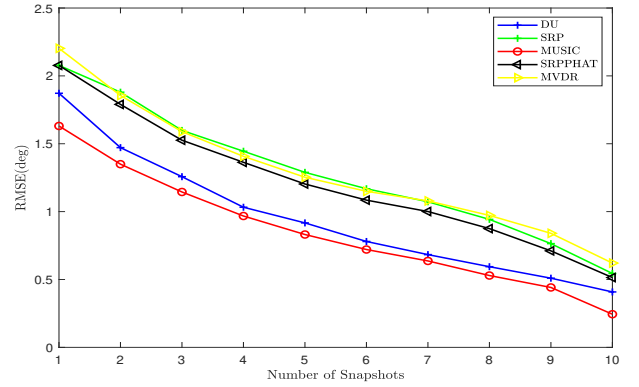


Fig. 3. RMSE performance versus the Number of Snapshots

of SNR with $T = 5$ snapshots and a reverberation time of 0.0s, while Figure 3 plots the RMSEs of the algorithms as a function of the number of snapshots with a signal to noise ratio SNR = 5 [dB].

It is clear that as both the SNR and the number of Snapshots increase, the estimation of the Direction of Arrival significantly improved. Indeed, DU demonstrates comparable performance to that of MUSIC, while also providing a lower level of complexity. When considering an SNR below 0 dB, it becomes evident that the algorithms exhibit a noticeable increase in RMSE. This result is attributed to the alignment of the noise subspace with larger eigenvalues. Conversely, as the SNR surpasses 0 dB, it becomes apparent that the RMSE for both the DU and MUSIC methods consistently remains below 1 degree. This favorable outcome is a result of our capacity to accurately estimate the noise subspace.

To assess the localization effectiveness of the DU algorithm in reverberant environment, we conducted a performance study of this algorithm based on the reverberation time RT60. Figure 4 illustrates the relationship between RMSE and RT60, demonstrating how the accuracy of the DU algorithm varies according to the reverberation times.

As depicted in Figure 4, the DU algorithm shows a level of performance comparing with that of the MUSIC algorithm. Furthermore, it is worth noting that the DU algorithm exhibits

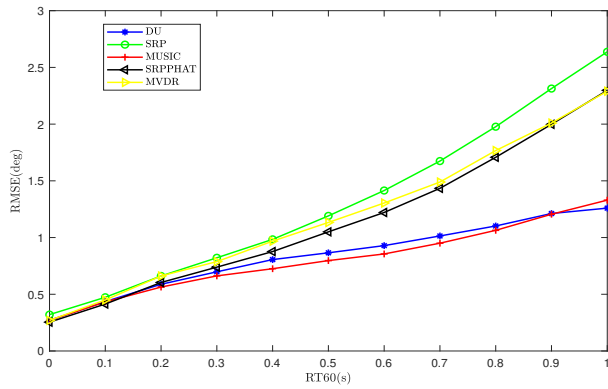


Fig. 4. RMSE performance versus RT60

superior performance compared to the MVDR and SRP PHAT techniques, which is acknowledged for its effectiveness in dealing with reverberant environments.

IV. EXPERIMENTAL RESULTS

In this section, we experimentally validate the performance of the DOA estimation algorithms using captured acoustic drone signal from a circular microphone array 8 microphones in indoor environments. The experiments are performed using Gras 46BE microphones and two Data Translation 9837B acquisition card, each of these cards is equipped with four analog inputs, we used a flying drone as an acoustic source. The sampling rate is set to $44.1kHz$, the number of snapshots is $T = 5$, the sound speed is $c = 343m/s$ and radius of the array is fixed to $r = 5cm$. The experimental platform is shown in Figure 5.

The results presented in Figure 6 illustrate that the microphone array successfully achieved high-precision localization of the drone, without the presence of secondary lobes that could distort the results. Furthermore, it is noteworthy that the localization peak is narrow, indicating fine spatial resolution and an ability to accurately discriminate the position of the mini-drone.

V. CONCLUSION

In this paper, a circular geometric configuration consisting of eight microphones is employed, where the performance of the MVDR, MUSIC and DU algorithms are compared. The obtained results clearly demonstrated that the DU algorithm, with its low computational complexity, is a good choice for achieving high resolution while maintaining superior performance. Indeed, the performance of the DU algorithm are examined in comparison to other algorithms such as MUSIC, MVDR, SRP, and SRP-PHAT by varying SNR, the number of snapshots and the reverberation time RT60. Our results reveal that the DU algorithm produces results comparable to those of the MUSIC algorithm while requiring less computational complexity. These results are further validated in a real-world scenario where signals generated by drone propellers are used

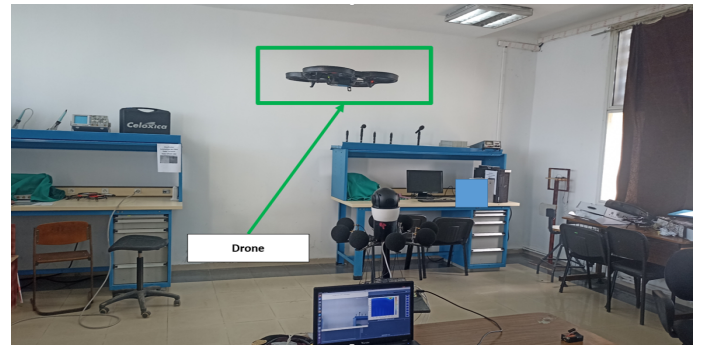


Fig. 5. Experimental platform.

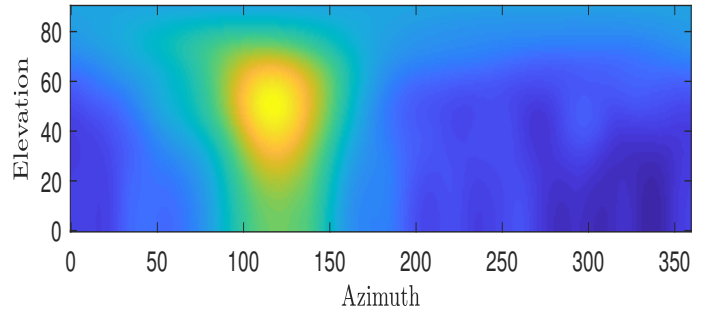


Fig. 6. Acoustic map.

and an acquisition platform with a circular geometric configuration that encompassed eight microphones is employed. In this practical setting, we achieved satisfactory results.

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